

Energy-Based Interpretation of General Relativity

Author: Andrew Robus

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1. Starting Point of General Relativity

General Relativity describes gravitation geometrically. Matter and energy are described by the energy-momentum tensor $T_{\mu\nu}$.

The Einstein field equations read:

- $G_{\mu\nu} = (8 * \pi * G / c^4) * T_{\mu\nu}$

Where:

- $G_{\mu\nu}$ describes the geometry and curvature of spacetime,
- $T_{\mu\nu}$ describes the local distribution of energy, momentum, pressure, and stress.

The geometry, in turn, determines the motion of energy and matter. This establishes the well-known feedback loop:

- $T_{\mu\nu} \rightarrow g_{\mu\nu} \rightarrow \text{dynamics} \rightarrow T_{\mu\nu}$

2. Our Fundamental Modification

The hypothesis initially does not introduce a new theory of gravitation. Instead, the ontological distinction between:

- matter,
- radiation,
- field energy,
- and other forms of energy

is abandoned at the fundamental level.

The core assumption is:

- $\text{Universe} = E$

The universe fundamentally consists of waves emitted from a central white hole at its center. Each wave consists of a chain of sequentially connected photons. It is highly plausible that each wave is also connected to its neighboring waves, forming a 3-dimensional photon field rather than individual, disconnected waves.

In this interpretation, "matter" is not an independent building block. The fundamental wave structure consists of photons; matter arises as a stable, organized state of this wave structure.

3. What Does "Bound Energy" Mean?

Within this hypothesis, the term does not describe an additional fundamental energy or matter component. It denotes an energy state in which the originally freely propagating wave structure organizes itself through interactions into a stable or at least long-lived, localized structure.

Such binding does not necessarily need to be explained by an additional fundamental force. Within General Relativity, every energy distribution contributes to the energy-momentum tensor and thereby generates spacetime curvature. A sufficiently concentrated and appropriately organized energy distribution can therefore, in principle, influence its own gravitational environment, which in turn feeds back onto the underlying wave structure.

This gives rise to an open physical question: whether the electromagnetic energy of the fundamental wave structure can form self-organized, stable, or at least long-lived states through its own energy-momentum distribution and the resulting spacetime curvature.

In simplified terms, the following development could emerge:

- **free wave energy** → **local interaction** → **oscillation / polarization** → **energy concentration** → **gravitational feedback** → **stable bound energy state**

The individual photons would not acquire any rest mass in this process. Crucially, the entirety of the energy participating in the stable structure forms a state with a non-zero rest energy.

Whether such self-organization and gravitational self-binding can actually lead to stable physical states is a further mathematical question. In particular, it would require a joint examination of the Einstein equations and electromagnetic dynamics.

For a bound state with rest energy, the following applies:

- $E_0 = m \cdot c^2$

The inertia associated with it is therefore not viewed within this hypothesis as an additional fundamental property alongside energy, but as an expression of the respective energy state. Mass (m) is accordingly not interpreted as its own fundamental building block, but as an equivalent description of a specific form of rest energy:

- $m = E_0 / c^2$

4. The Energy-Momentum Tensor Remains Intact

The crucial point is: we do not replace $T_{\mu\nu}$. On the contrary, it becomes the central mathematical tool for describing the fundamental energy.

- $T_{\mu\nu}$ = **local physical state of energy**

It continues to contain, in particular:

- energy density,
- momentum density,
- energy flux,
- pressure,

- stresses.

Thus, no new "matter equation" needs to be introduced for matter. A matter system continues to be described by its corresponding $T_{\mu\nu}$.

Our modification lies in stating: matter is ontologically not an additional fundamental constituent. It is a bound, organized state of the underlying energy and wave structure. The energy and momentum distribution of this state that is relevant to spacetime is described by the energy-momentum tensor.

5. The Einstein Field Equations Remain Unchanged

Consequently:

- $G_{\mu\nu} = (8 \cdot \pi \cdot G / c^4) \cdot T_{\mu\nu}$

remains fully intact. The change on the right-hand side is not a new formula, but a change in its interpretation.

- **Standard interpretation (simplified):** Matter / Energy $\rightarrow T_{\mu\nu}$
- **Our interpretation:** Energy $\rightarrow T_{\mu\nu}$

Matter is merely a special state of this energy.

6. Energy and Geometry

Within this hypothesis, the following chain emerges:

- $E \rightarrow T_{\mu\nu} \rightarrow G_{\mu\nu} \rightarrow g_{\mu\nu}$

Energy thus determines geometry via its energy-momentum state. Conversely, geometry determines the possible development of energy and its states of motion. This preserves the mutual feedback loop of GR:

- $T_{\mu\nu} \leftrightarrow g_{\mu\nu}$

7. Energy Can Change Its State

Within this hypothesis, fundamental energy consists of the described photonic wave (field) structure. Energy itself is not split into fundamentally distinct categories such as "free" and "bound" energy. Rather, the decisive factor is the state and spatial organization of the wave structure.

Local interactions can cause the structure to change. Initially freely propagating waves can form stable or long-lived oscillation and organization systems. Conversely, such stable structures can change state again, transferring energy into other wave states.

Schematically, this process can no longer be described as:

- $E_{\text{bound}} \rightarrow E_{\text{free}}$

but rather, for example, as:

- **wave structure $\rightarrow$ interaction $\rightarrow$ stable oscillation structure $\rightarrow$ state change / energy transfer $\rightarrow$ new wave structure**

The underlying energy is conserved throughout. What changes is its distribution, form of motion, oscillation structure, and organization within the overall system.

Electromagnetic radiation in this interpretation would not be the creation of a fundamentally different kind of energy, but a transfer or reorganization of the existing photonic wave structure from a stable structure into the surrounding space.

For the entire closed universe, local energy conservation in the form of a continuity equation still holds schematically:

- **(partial derivative of energy density ρ_E with respect to time t) + divergence of energy flux vector $S_E = 0$**

Where ρ_E describes the local energy density and S_E describes the local energy flux.

The core statement is therefore: energy is neither created nor destroyed within this hypothesis, but changes its spatial distribution and structural state.

8. The Universe as a Single Total System

The hypothesis further assumes that there is no external medium and no external space into which the universe expands. Instead, the universe is understood as a single connected physical total system.

The initial state is assumed to be an extremely concentrated and high-energy wave structure:

- **energy density $\rho_E(t_0, x) = \text{extremely high energy density}$**

The cosmic evolution is subsequently the temporal and spatial development of this energy and wave structure:

- **State $\Psi(t_0) \rightarrow \text{State } \Psi(t)$**

The fundamental nature of energy does not change during this process. Rather, different states of organization and development of the underlying photonic wave structure arise through interactions, propagation, oscillations, and feedback.

Schematically:

- **fundamental wave structure $\rightarrow$ interaction $\rightarrow$ different oscillation and organization states $\rightarrow$ stable structures / propagating states $\rightarrow$ further state changes**

Matter and radiation therefore do not represent ontologically distinct basic constituents within this interpretation. They are different physical states or organizational forms of the same fundamental system.

9. What Actually Changes in GR?

This can be stated very concisely:

Unchanged:

- The field equation $G_{\mu\nu} = (8 \cdot \pi \cdot G / c^4) \cdot T_{\mu\nu}$ remains intact.
- The geometric description of gravity, covariant energy-momentum conservation, and geodesic motion are also initially retained.

Changed Ontology:

- Instead of viewing matter, radiation, and fields as distinct fundamental physical contents, the hypothesis states: **Everything is an expression of a single fundamental energy and wave content** (or photon field content, where, due to the polarization of the individual straight-line waves, it is assumed to consist of individual waves lying so close together that they collectively act as a field).
- Matter is understood as a stable or long-lived, organized state of this energy and wave structure.
- Radiation, on the other hand, corresponds to a predominantly propagating state of the same fundamental structure.
- The various physical states are described regarding their gravitationally relevant properties by their respective energy-momentum state $T_{\mu\nu}$.

10. The Central Formula of the Hypothesis

Universe = E

with

- $E \rightarrow T_{\mu\nu}$

and

- $G_{\mu\nu} = (8 * \pi * G / c^4) * T_{\mu\nu}$

Combined:

- **Universe = E $\rightarrow$ $T_{\mu\nu}$ $\rightarrow$ $G_{\mu\nu}$ $\rightarrow$ $g_{\mu\nu}$ $\rightarrow$ dynamics of E**

The feedback closes the loop:

- **E $\rightarrow$ $T_{\mu\nu}$ $\rightarrow$ $g_{\mu\nu}$ $\rightarrow$ E**

The core modification compared to the standard interpretation of GR does not lie in a new Einstein equation, but in the ontological baseline assumption:

The various physical manifestations are not treated as independent fundamental components of reality. Instead, it is assumed that they represent different states, organizational forms, and dynamics of a single fundamental energy and wave content.

Matter would thus not be an additional building block alongside energy, but a stable, organized state of the same fundamental structure.

The next scientific test consists of examining whether this interpretation can be consistently formulated with already known forms of $T_{\mu\nu}$ —such as dust, fluid, radiation, and electromagnetic fields—without altering the successful mathematical structure of General Relativity.

AI Assistance Statement

Generative AI tools, including ChatGPT (OpenAI), were used as research and writing assistants during the development of this hypothesis. Their use included discussion of physical concepts, critical examination of assumptions, organization and revision of the manuscript, and language editing.

The author remains solely responsible for the scientific content, assumptions, interpretations and conclusions presented in this work. AI-generated suggestions were critically evaluated and do not constitute independent scientific validation of the hypothesis.